Class 11: AlphaFold

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Background

We saw last day that the main repository for biomolecular strcuture (the PDB database) only has ~250,000 entries.

UniProtKB (the main protein sequence database) has over 200 million entries!

AlphaFold

In this hands on session we will utilize AlphaFold to predict protein structure from sequence (Jumper et al. 2021).

Without the aid of such approaches, it can take years of expensive lab work to determine the structure of just one protien. With ALphaFold we can now accurately compute a typical protein structure in as little as 10 minutes.

The EBI AlphaFold database

The EBI AlphaFold database contains lots of computed structure models. It is increasingly likely that the structure you are interested in is already in the database.

There are 3 major outputs from AlphaFold.

1. A model of structure in **PDB** format.
2. A **plDDT score** that tells us how confident the model is for a given residue in your protein (high values are good, above 70).
3. A **PAE score** that tells us about protein packing quality.

If you can't find a matching entry for the sequence you are interested in AFDB you can run AlphaFold yourself…

Running AlphaFold

We will use ColabFold to run AlphaFold on our sequence https://colab.research.google.com/github/sokrypton/ColabFold/blob/main/AlphaFold2.ipynb

Interpreting Results

Custom analysis of resulting models

We can read all the AlphaFold results into R and do more quantitative analysis than just viewing the structures in Mol-star:

Read all the PDB models:

```
library(bio3d)
p <- read.pdb("hivpr_23119/hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed
```

```
pdb_files <- list.files("hivpr_23119/", pattern=".pdb", full.names=TRUE)
```

```
pdbs <- pdbaln(pdb_files, fit=TRUE, exefile="msa")
```

```
Reading PDB files:
hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2_seed_000.pdb
hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3_seed_000.pdb
```

```
.....

Extracting sequences

pdb/seq: 1   name: hivpr_23119//hivpr_23119_unrelaxed_rank_001_alphafold2_multimer_v3_model_4
pdb/seq: 2   name: hivpr_23119//hivpr_23119_unrelaxed_rank_002_alphafold2_multimer_v3_model_1
pdb/seq: 3   name: hivpr_23119//hivpr_23119_unrelaxed_rank_003_alphafold2_multimer_v3_model_5
pdb/seq: 4   name: hivpr_23119//hivpr_23119_unrelaxed_rank_004_alphafold2_multimer_v3_model_2
pdb/seq: 5   name: hivpr_23119//hivpr_23119_unrelaxed_rank_005_alphafold2_multimer_v3_model_3
```

How similar or different are my models?

```r
rd <- rmsd(pdbs, fit=T)
```

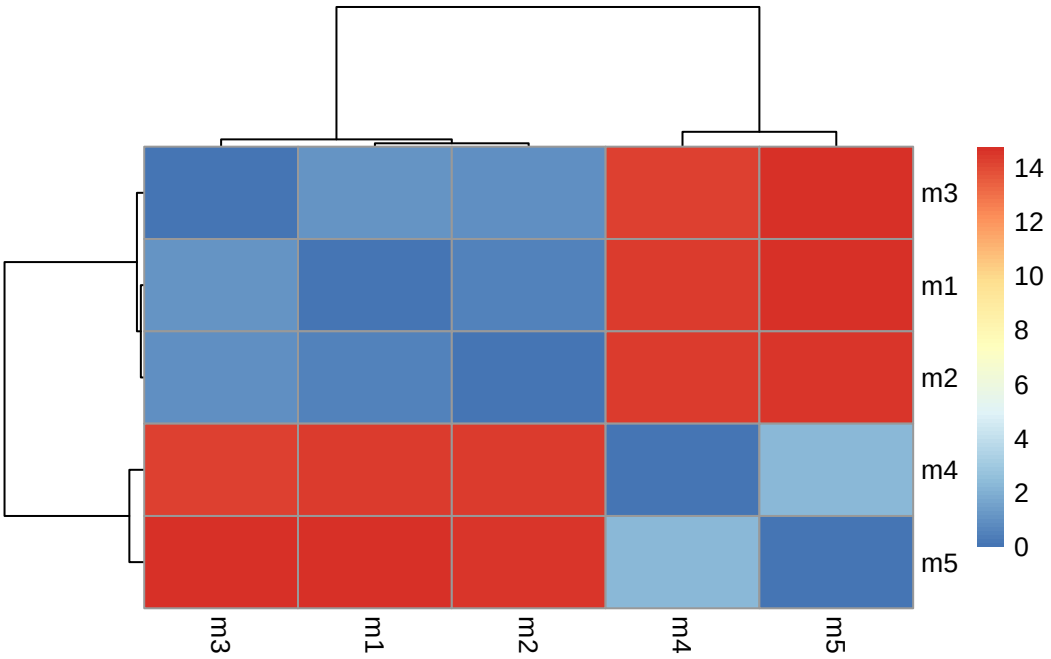
```
Warning in rmsd(pdbs, fit = T): No indices provided, using the 198 non NA positions
```

```r
range(rd)
```

```
[1]  0.000 14.754
```

```r
library(pheatmap)

colnames(rd) <- paste0("m",1:5)
rownames(rd) <- paste0("m",1:5)
pheatmap(rd)
```

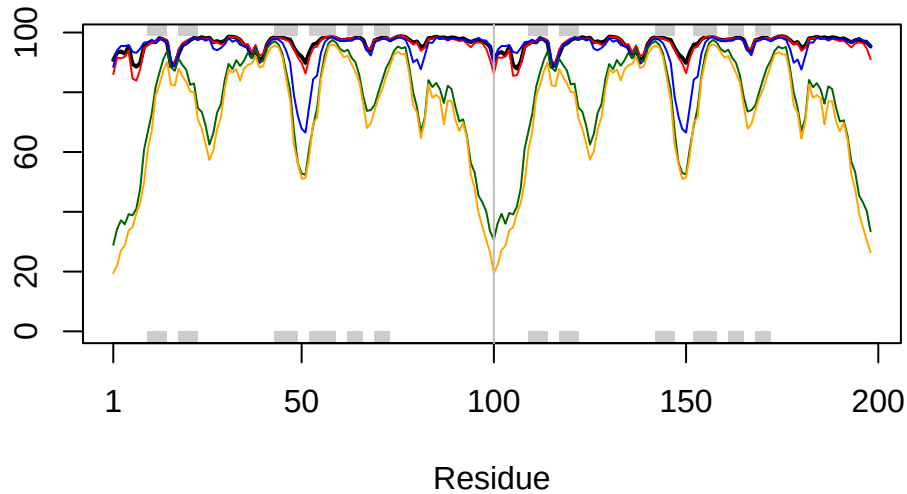


Plot pLDDT values across all models:

```
# Read a reference PDB structure
pdb <- read.pdb("1hsg")
```

```
  Note: Accessing on-line PDB file
```

```
plotb3(pdbs$b[1,], typ="l", lwd=2, sse=pdb)
points(pdbs$b[2,], typ="l", col="red")
points(pdbs$b[3,], typ="l", col="blue")
points(pdbs$b[4,], typ="l", col="darkgreen")
points(pdbs$b[5,], typ="l", col="orange")
abline(v=100, col="gray")
```



Improving the superposition/fitting of our models:

```
core <- core.find(pdbs)
```

```
core size 197 of 198  vol = 9885.419
core size 196 of 198  vol = 6898.241
core size 195 of 198  vol = 1338.035
core size 194 of 198  vol = 1040.677
core size 193 of 198  vol = 951.865
core size 192 of 198  vol = 899.087
core size 191 of 198  vol = 834.733
core size 190 of 198  vol = 771.342
core size 189 of 198  vol = 733.069
core size 188 of 198  vol = 697.285
core size 187 of 198  vol = 659.748
core size 186 of 198  vol = 625.28
core size 185 of 198  vol = 589.548
core size 184 of 198  vol = 568.261
core size 183 of 198  vol = 545.022
core size 182 of 198  vol = 512.897
core size 181 of 198  vol = 490.731
core size 180 of 198  vol = 470.274
core size 179 of 198  vol = 450.738
```

```
core size 178 of 198  vol = 434.743
core size 177 of 198  vol = 420.345
core size 176 of 198  vol = 406.666
core size 175 of 198  vol = 393.341
core size 174 of 198  vol = 382.402
core size 173 of 198  vol = 372.866
core size 172 of 198  vol = 357.001
core size 171 of 198  vol = 346.576
core size 170 of 198  vol = 337.454
core size 169 of 198  vol = 326.668
core size 168 of 198  vol = 314.959
core size 167 of 198  vol = 304.136
core size 166 of 198  vol = 294.561
core size 165 of 198  vol = 285.658
core size 164 of 198  vol = 278.893
core size 163 of 198  vol = 266.773
core size 162 of 198  vol = 259.003
core size 161 of 198  vol = 247.731
core size 160 of 198  vol = 239.849
core size 159 of 198  vol = 234.973
core size 158 of 198  vol = 230.071
core size 157 of 198  vol = 221.995
core size 156 of 198  vol = 215.629
core size 155 of 198  vol = 206.8
core size 154 of 198  vol = 196.992
core size 153 of 198  vol = 188.547
core size 152 of 198  vol = 182.27
core size 151 of 198  vol = 176.961
core size 150 of 198  vol = 170.72
core size 149 of 198  vol = 166.128
core size 148 of 198  vol = 159.805
core size 147 of 198  vol = 153.775
core size 146 of 198  vol = 149.101
core size 145 of 198  vol = 143.664
core size 144 of 198  vol = 137.145
core size 143 of 198  vol = 132.523
core size 142 of 198  vol = 127.237
core size 141 of 198  vol = 121.579
core size 140 of 198  vol = 116.78
core size 139 of 198  vol = 112.575
core size 138 of 198  vol = 108.175
core size 137 of 198  vol = 105.137
core size 136 of 198  vol = 101.254
```

```
core size 135 of 198  vol = 97.379
core size 134 of 198  vol = 92.978
core size 133 of 198  vol = 88.188
core size 132 of 198  vol = 84.032
core size 131 of 198  vol = 81.902
core size 130 of 198  vol = 78.023
core size 129 of 198  vol = 75.276
core size 128 of 198  vol = 73.057
core size 127 of 198  vol = 70.699
core size 126 of 198  vol = 68.976
core size 125 of 198  vol = 66.707
core size 124 of 198  vol = 64.376
core size 123 of 198  vol = 61.145
core size 122 of 198  vol = 59.029
core size 121 of 198  vol = 56.625
core size 120 of 198  vol = 54.369
core size 119 of 198  vol = 51.826
core size 118 of 198  vol = 49.651
core size 117 of 198  vol = 48.19
core size 116 of 198  vol = 46.644
core size 115 of 198  vol = 44.748
core size 114 of 198  vol = 43.288
core size 113 of 198  vol = 41.089
core size 112 of 198  vol = 39.143
core size 111 of 198  vol = 36.468
core size 110 of 198  vol = 34.114
core size 109 of 198  vol = 31.467
core size 108 of 198  vol = 29.445
core size 107 of 198  vol = 27.323
core size 106 of 198  vol = 25.82
core size 105 of 198  vol = 24.149
core size 104 of 198  vol = 22.647
core size 103 of 198  vol = 21.068
core size 102 of 198  vol = 19.953
core size 101 of 198  vol = 18.3
core size 100 of 198  vol = 15.723
core size 99 of 198  vol = 14.841
core size 98 of 198  vol = 11.646
core size 97 of 198  vol = 9.435
core size 96 of 198  vol = 7.354
core size 95 of 198  vol = 6.181
core size 94 of 198  vol = 5.667
core size 93 of 198  vol = 4.706
```

```
 core size 92 of 198  vol = 3.664
 core size 91 of 198  vol = 2.77
 core size 90 of 198  vol = 2.151
 core size 89 of 198  vol = 1.715
 core size 88 of 198  vol = 1.15
 core size 87 of 198  vol = 0.874
 core size 86 of 198  vol = 0.685
 core size 85 of 198  vol = 0.528
 core size 84 of 198  vol = 0.37
 FINISHED: Min vol ( 0.5 ) reached
```

```r
core.inds <- print(core, vol=0.5)
```

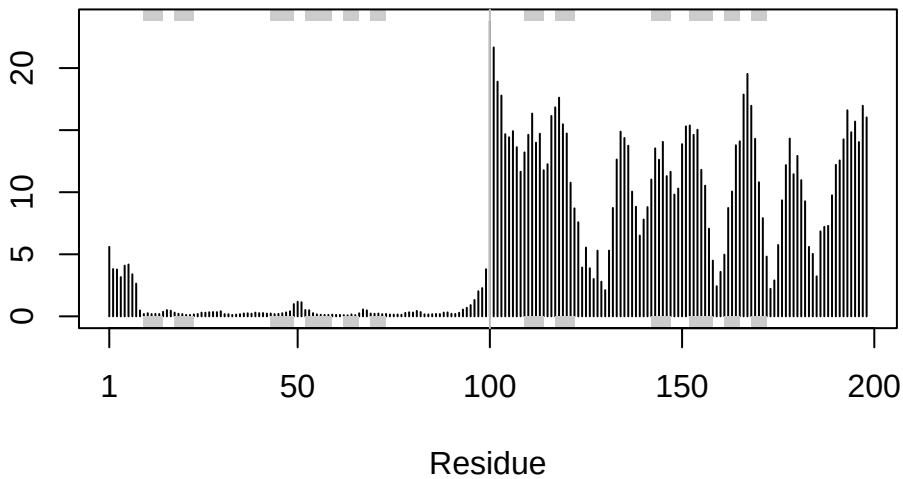
```
# 85 positions (cumulative volume <= 0.5 Angstrom^3)
  start end length
1     9  49     41
2    52  95     44
```

```r
xyz <- pdbfit(pdbs, core.inds, outpath="corefit_structures")
```

Examining the RMSF between positions of the structure. RMSF is used as a measure of conformational variance along the structure:

```r
rf <- rmsf(xyz)

plotb3(rf, sse=pdb)
abline(v=100, col="gray", ylab="RMSF")
```



We can see that the first chain is very similar across different models, but the second chain is more variable, which was also visualized in Mol.

Predicted Alignment Error for Domains

AlphaFold produces an output called **Predicted Aligned Error (PAE)** which is independent of the 3D structure.

```
library(jsonlite)
# Listing of all PAE JSON files
pae_files <- list.files("hivpr_23119/",
                        pattern=".*model.*\\.json",
                        full.names = TRUE)
pae1 <- read_json(pae_files[1],simplifyVector = TRUE)
pae5 <- read_json(pae_files[5],simplifyVector = TRUE)

attributes(pae1)
```

```
$names
[1] "plddt"   "max_pae" "pae"     "ptm"     "iptm"
```

```r
# Per-residue pLDDT scores
#  same as B-factor of PDB..
head(pae1$plddt)
```

```
[1] 90.81 93.25 93.69 92.88 95.25 89.44
```

```r
pae1$max_pae
```

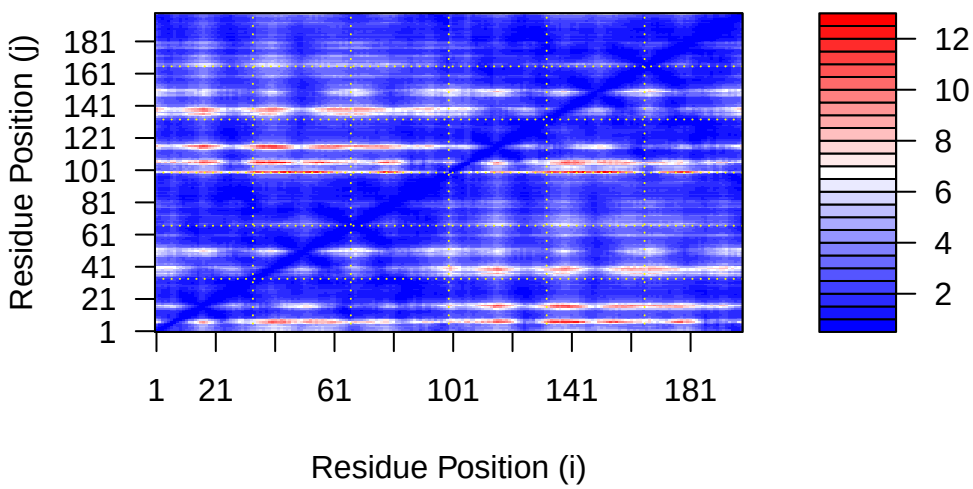
```
[1] 12.84375
```

```r
pae5$max_pae
```

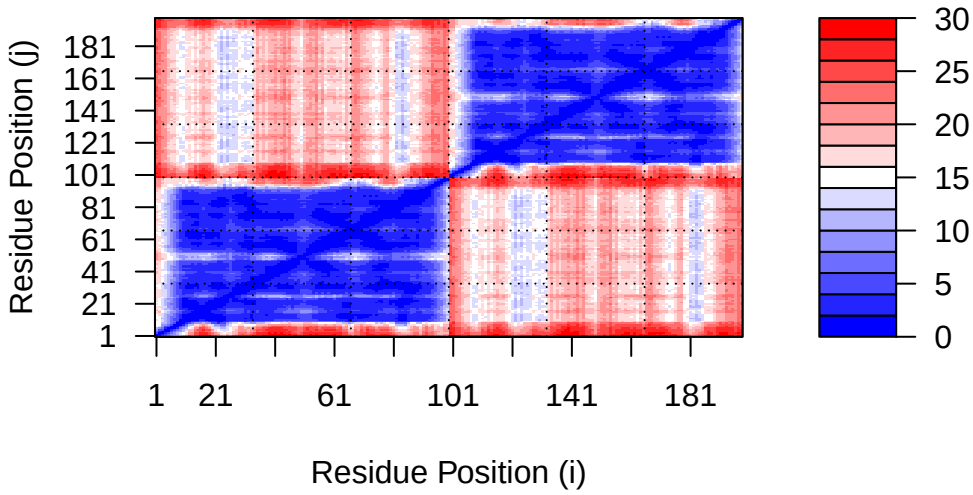
```
[1] 29.59375
```

We can plot N by N (where N is the number of residues):

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)")
```

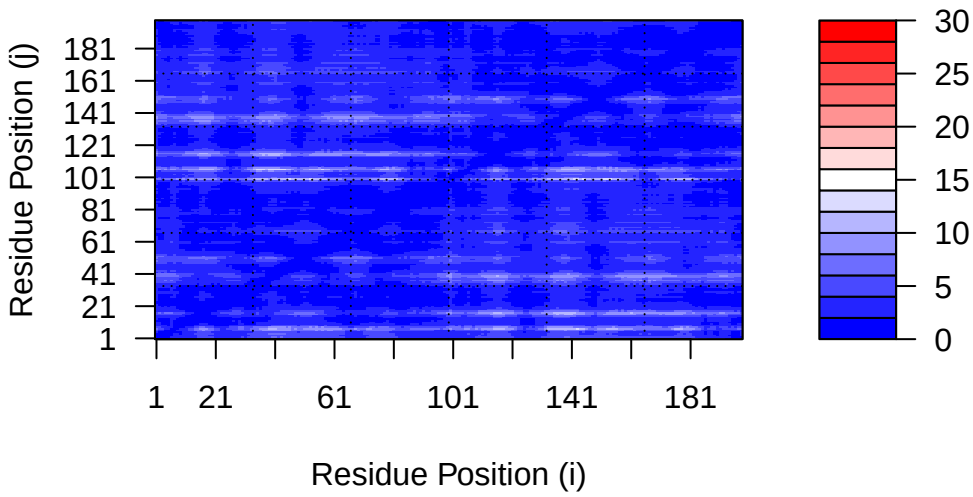


```r
plot.dmat(pae5$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Same plot, but using the same data range as the plot for model 5:

```r
plot.dmat(pae1$pae,
          xlab="Residue Position (i)",
          ylab="Residue Position (j)",
          grid.col = "black",
          zlim=c(0,30))
```



Residue Conservation from Alignment File

```
aln_file <- list.files("hivpr_23119/",
                       pattern=".a3m$",
                        full.names = TRUE)
aln_file
```

```
[1] "hivpr_23119//hivpr_23119.a3m"
```

```
aln <- read.fasta(aln_file[1], to.upper = TRUE)
```

```
[1] " ** Duplicated sequence id's: 101 **"
[2] " ** Duplicated sequence id's: 101 **"
```

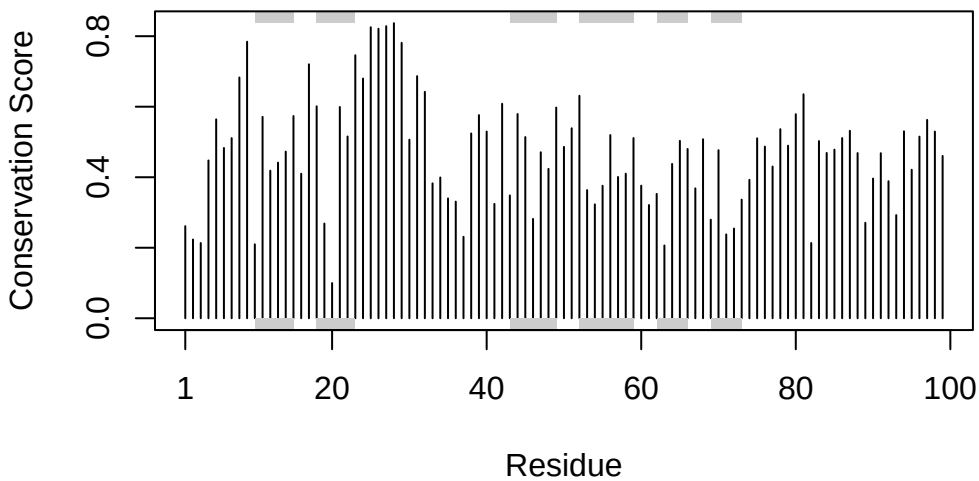
How many sequences are in this alignment:

```
dim(aln$ali)
```

```
[1] 5397  132
```

Scoring residue conservation in the alignment with the `conserv()` function.

```
sim <- conserv(aln)
plotb3(sim[1:99], sse=trim.pdb(pdb, chain="A"),
       ylab="Conservation Score")
```



Note the conserved Active Site residues D25, T26, G27, A28, which we can see if we generate a consensus sequence with a high cutoff value:

```
con <- consensus(aln, cutoff = 0.9)
con$seq
```

```
  [1] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [19] "-" "-" "-" "-" "-" "-" "D" "T" "G" "A" "-" "-" "-" "-" "-" "-" "-" "-"
 [37] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [55] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [73] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
 [91] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[109] "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-" "-"
[127] "-" "-" "-" "-" "-" "-"
```

We can view these functionally important sites by mapping the conservation score to the Occupancy column of a PDB file to view in programs like MolStar.

```r
m1.pdb <- read.pdb(pdb_files[1])
occ <- vec2resno(c(sim[1:99], sim[1:99]), m1.pdb$atom$resno)
write.pdb(m1.pdb, o=occ, file="m1_conserv.pdb")
```

Current Limitations and Potential Problems

1. Multimer vs single chain predictiions: doesn't always give reasonable results.
2. Predicting different conformational states: can only predict a single state and is hard to tell which state of a protein will be captured.
3. Relative positions of domains: if a predicted model has more than one domain, each domain may have high confidence, but the relative positions of the domains may not.
4. Poorly predicted regions/bugs: there are protein regions that AlphaFold cannot predict.

Summary

AlphaFold provides biologists with a new technique of structure-gazing without the experimental structure determination work. As the next generation of biologists, it is important we learn how to critically analyze predicted structures, notice new interactions, and examine proteins of interest in sufficient detail to differentiate between what are "bugs" and what are "features".